

APPROVED



MSc Robotics
School of Computing, Engineering and Technology

Version	4.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	
No Course Award Assigned	
Code	0685
SCQF Level	Level 11
QAA Subject Benchmark	Engineering (2023)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100170	mechatronics and robotics	mechatronics and robotics	100%
Total			100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The MSc Robotics course aims providing an in-depth understanding of practical and theoretical knowledge of modern robotics technologies in order to develop skilled robotics engineers to support the various sector (automation, healthcare, manufacturing, energy, subsea, service, logistics, agriculture, space, military, education, transportation etc.) demands by 2030. The course is delivered full time through blended learning via the University's Moodle. The course involves the completion of eight study modules followed by an individual MSc Project. Each module is assessed by coursework assignments, and students are required to pass the 8 taught modules before progressing to the dissertation. Modules may be supplemented by industrial visits and industry speakers where practical and appropriate. Blended
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What the Course Involves	learning will be at the forefront to allow extensive use of RGU's Virtual Learning Environment (Moodle) which facilitates interaction with teaching staff and other students for full time delivery. Full Time Students • For September entry, the sequence of modules will be: Semester 1 (taken in September), Semester 2 (taken in January), Semester 3 (project, taken in May) • For January entry, the sequence of modules will be: Semester 2 (taken in January), Semester 1 (taken in September), Semester 3 (project, taken in January) Students are expected to attend on campus for the duration of the course. Full time mode of delivery is completed over a period of 12 calendar months, four modules per semester, followed by the individual project dissertation. Each student will be assigned an experienced academic supervisor to assist with the process of preparing for submission. The taught element of the course comprises four 15 credit modules in Semester 1 and four 15 credit modules in semester 2. The taught element of the course is followed, in Semester 3, by the 60 credit MSc Project. The MSc Project is a core module, central to the objectives of the course, and enables the student to put into practice the theories taught elsewhere on the course.
Course Collaborations	
No collaborations listed	
Introduction to the Course	This MSc Robotics course is an interdisciplinary course in the School of Computing, Engineering and Technology. The vision of this MSc Robotics course is to develop experts in robotics, autonomous system and AI technologies who can lead the future advancements in the domain. The main highlight of this MSc course is a project-based learning approach developed by our expert academics to enhance the integration of our engineers into the industry of the future. The course will provide graduates with practical experience as it incorporates practical activities, site visits, case study learning, and industry guest lectures. There will be an opportunity for completing the final project with industry. Final projects will also be supported by our School research groups working in various interdisciplinary areas, thus providing graduates with research skills as well. The course also includes group projects on real-life applications that address various aspects of the UN Sustainable Development Goals (UNSDGs). The key features of this MSc Robotics course will make the prospective students highly employable and entrepreneurial. This course is delivered as blended learning full-time.
Educational Aims of the Course	<u>This course aims to:</u> Develop skilled robotics engineers that can meet the expectations and increased needs of various industries including automation, healthcare, manufacturing, energy, subsea, service, logistics, agriculture, space, military, education, transportation etc. The aim of this course is to: • Produce fit-for-purpose engineers with professional skills in the key areas of robotics and Intelligent systems design, maintenance, and operational management. • Provide students with real practical and technical skills which enhance their career opportunities, typically in various industries, engineering firms and engineering service companies. • Provide students with independent investigation and research skills as well as presentation skills. Postgraduate Certificate Stage: Students learn the fundamentals and advances made in the technologies used within robotics and intelligent systems. Postgraduate Diploma Stage: Students learn how to design and develop robotic systems and control them while considering innovation and sustainability. Masters Stage: Students work on individual projects demonstrating their understanding and expertise of implementing the knowledge gained from Semester 1 and Semester 2 teachings into their project.

Awarding Institution	Robert Gordon University
Course Duration	Full-Time Students Full-time 12 months with a September intake Full-time 16 months with a January intake Students can choose to undertake a short-term (3 month) placement or a long-term (12 months) placement. All placements require approval from the School prior to commencement. If a student goes on placement, the duration of the course will be extended by the duration of the placement.
Distinctive Features of the Course	The course will offer students a unique opportunity to gain in-depth knowledge and experience in robotics engineering in addition to skills required by industry such as project management and experimental work. Students will graduate with robotic system/solution design and integration capabilities including AI, machine learning and machine vision skills, which are required by many companies. By successfully completing the MSc Robotics, graduates will have the advantage over other graduates from other robotics courses to apply for jobs due to their unique practical and research skills attained. Examples of jobs include: • Robotics engineering project manager • Robotics design engineer • AI and Machine learning specialist • Robotics Project engineer • Robotics equipment manufacturers • Engineering and consulting specialist in various industrial sectors including automation, healthcare, manufacturing, energy, subsea, services, logistics, agriculture, space, military, education, transportation etc. The teaching of this course will involve gaining research expertise and research outputs will form part of the course in time. <u>Students will benefit from:</u> • On-Campus robotics lab, ROV pool facility • Expert Guest Lectures • Practical Activities • Final Project Industrial Placement Opportunities through RGU established Collaboration with various engineering Companies.
	Students may elect to undertake a short placement or a year-long placement. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement.

Placement/Professional Experience	While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short placement. This normally comprises 3 months of continuous, full-time supervised work experience. The assessment for a short placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short placement will normally receive at least 1 video call with their Academic Tutor. As an alternative, students may undertake a year-long placement. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. Placements must be undertaken after the completion of all taught modules and begin before the start of the final project. A student may choose to either undertake a placement at the same time as their final project or delay the start of their final project until the placement has finished. The course leader must approve this decision. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 180 credits achievable from the other modules. Successful completion is not required for progression or graduation. The credits achieved for placement may not be used to compensate for any credits from the other modules.
Opportunities for Further Study	Following the completion of this MSc, further study of Doctor of Philosophy (PhD) or Engineering Doctorate (EngD) is possible.
External and Internal Indicators	Internal references include: • University Mission Statement External references include: • QAA Subject Benchmark for Computing and Engineering • SCQF Qualification Descriptors, Level 11 • Suitable appropriate Accreditation (planned for after first delivery of the course)

of Quality and Standards	• Peer review of staff research RGU's School of Computing, Engineering and Technology provides Industry with highly qualified and committed graduates.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Applicants normally hold a 2:2 undergraduate honours degree, or its equivalent in Engineering or relevant Science discipline. Applicants whose first language is not English should have an IELTS of 6.5 overall, with no component lower than 5.5, or an equivalent.
Alternative Qualifications for Entry	Applicants with a lower degree classification and/or relevant work experience will be considered on a case by case basis.
Interview Procedures	Applicants are not usually interviewed, however, some may be interviewed if they do not meet the standard entry requirements.
Course-Specific Regulations	Module ENM409 cannot be condoned. Students are awarded PgCert Robotics on successful completion of 60 credits at SCQF Level 11 with at least 15 credits from successful completion of either ENM280 Advanced Robotics or ENM286 Interdisciplinary Robotics Group Project.

	All students, regardless of mode and start date, will typically undertake resits in the summer exam period. Exceptionally students who start in January may be offered a resit opportunity after completing two semesters, in April of the following year.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region.

	RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	The MSc Robotics is designed to allow students to acquire specialist knowledge and generic skills in engineering, political and economic contexts of various industrial sectors including automation, healthcare, manufacturing, energy, subsea, service, logistics, agriculture, space, military, education, transportation etc. Graduates from the course have a greater advantage when applying for robotics, automation and AI/ML jobs compared to graduates from the more generalised engineering courses. The MSc Robotics will offer the student a unique experience and opportunity to gain in-depth knowledge in various robotics, automation, autonomous and AI/ML systems in addition to industrial skills required by the industry such as project management and experimental study. The course is suitable for graduates from a range of engineering and related disciplines and has been specifically designed to meet the various high-growth global industrial sectors related to automation, healthcare, manufacturing, energy, subsea, service, logistics, agriculture, space, military, education, transportation etc. Students on this course are able to draw on a range of high-quality resources specific to robotics, such as a wide range of RGU Library resources including databases such as Web of Science, extensive journal subscriptions and a wide range of robotics related textbooks and e-books. The staff team is able to draw on their active experience with research and consultancy to provide teaching, which is relevant, up-to-date, and informed by real world experience. This course will enable the students to critically apply their thinking to effectively examine the theories and practices that make up the professional knowledge in robotics. Building on the students' first degree and the experience students have gained since graduating, this MSc course will allow the students to develop more advanced skills and knowledge in the field of robotics. The School of Computing, Engineering and Technology, and RGU have close links to a number of representatives within the local, national and international companies, and local representatives from industry organisations. The course is delivered in single mode, Blended Learning Full-Time via Moodle. Lectures and tutorials are combined with practical or computer-based simulation activities of industrial relevance. Students will also have guest speaker lecturers from industry representatives specialised in different fields of robotics. Students will also benefit from our exclusive robotics lab and fully equipped IT labs.
Teaching and Learning Strategy	The curriculum of the MSc robotics is designed to enable students to acquire and develop their subject knowledge and understanding, intellectual and thinking ability and practical and key transferable skills at a Masters level corresponding to the Scottish Credit and Qualifications Framework. The Teaching and Learning Strategy for this course aligns with the School's Strategy as listed below. In line with the University' Teaching & Learning Strategy, the School of Computing, Engineering and Technology: 1. Aims to produce industrially relevant, professional engineering managers. 2. Encourages all staff in the development of innovative and effective teaching methods. 3. Believes that strong interaction with industry is important in enhancing the Teaching & Learning process. 4. Seeks to develop students' abilities to work as part of a team. 5. Is committed to the use of feedback from students to enhance the Teaching & Learning process. 6. Is keen to develop its students as independent and enthusiastic learners. 7. Recognises the value of effective and timely feedback to students. 8. Places importance on the appropriate use of technology in Teaching and Learning. 9. Appreciates the importance of Personal Development Planning (PDP) and the development of transferrable skills. 10. Acknowledges that the development of good Teaching & Learning practices is a time-consuming/demanding activity. 11. Is keen to enhance the learning process for its students to encourage creativity and the development of problem-solving skills. 12. Recognises the importance of providing a high quality learning environment.

	Staff are involved in extending the use of a variety of teaching and assessment methods, including Active Learning, computer-based quizzes and the delivery of lecture-based material by means of the VLE (Moodle) to enhance the Teaching and Learning process.
Assessment	The assessment strategy for the course is guided by the course structure, which involves eight 15 credit modules in the Postgraduate Diploma stage, followed by the 60-credit individual MSc project. The eight modules in the Postgraduate Diploma stage are assessed by coursework. In this course, core of the various module contents is suitable for open-ended design work which is best suited to coursework. This is kept under review to ensure that students are assessed appropriately. The group project and individual project allows many elements of the course to be assessed. The project activity is assessed by a thesis to be submitted. Students are provided with a detailed project handbook which describes the various stages in the project and the means by which it is assessed. Knowledge and Understanding Knowledge and Understanding can be assessed by means of: • Computer-based assessments • Coursework projects written/oral assessment Some elements of knowledge and understanding are assessed by means of the MSc project written or oral assessment. Practice: Applied Knowledge, Skills and Understanding Practical skills are assessed by the most appropriate means, depending on the nature of the practical work. These skills are assessed by means of: laboratory exercises and reports, case studies, design exercises, and computer based assessment (e.g. simulation) project work. Similarly, a set of practical skills are also assessed in the MSc Robotics individual research Investigation (ENM409) which involves demonstration of practical skills required to complete a substantial practical robotics design task. Generic Cognitive Skills Assessment of Generic Cognitive Skills generally involves the production of technical report content which reflects on problem identification, critical appraisal of alternative solution strategies, evaluation of solution performance. The assessment of generic cognitive skills is particularly evident in the two MSc Project modules (ENM286 Interdisciplinary Robotics Group Project and ENM409 MSc Robotics Individual Research Investigation). Communication, ICT and Numeracy Skills Communication: Technical reports documenting robotic system design assignments; structured project documents to manage project progress, establish quality guidelines, archive project deliverables; use of campus resources to present MSc Project outcomes. Team communication tasks are implicitly assessed in group work tasks which involve both oral/written communication between team members including document sharing. ICT: The use of IT tools in all coursework assignments (i.e. use of tools to plan, aid design, collate results and document coursework). Demonstration of IT competence is mandatory in production of documentation to support all coursework submissions including development of support materials for the project oral presentation. Numeracy skills: these skills are essential for the successful completion of most course modules and are implicitly assessed in courseworks. Numeracy skills are particularly key to Advanced Robotics and Machine Learning data analysis tasks and the evaluation of project results. Autonomy, Accountability and Working with Others

	These skills are assessed implicitly in all modules. Explicit assessment of these skills is evident in the marking of the project, which reflects the student's interaction with their supervisor and includes demonstration of own-initiative and ability to work independently. Team-working skills are mainly assessed in the Interdisciplinary Robotics Group Project (ENM286) which includes group work while management of complex ethical and professional issues are assessed through coursework reports in Sustainable Enterprise development (ENM262) module which focuses on delivery of selected United Nation Sustainable Development Goal (UNSDG) innovation project.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The course is delivered full time on campus and all study materials will be available on the Virtual Learning Environment (Moodle). The University's Moodle facility enables easy access to a wide range of teaching resources. For tutorials, the cohort may be split into multiple groups to allow students support each other. Personal tutors are also available to students for support and guidance. Field visits allow practical aspects of module content to be more effectively illustrated than is possible in a lecture theatre or tutorial room environment, the students on the MSc Robotics therefore have the exclusive access to various robotics platforms in the RGU Robotics lab and the ROV pool facility to be used for real condition tests. The online material on Moodle and library, where applicable, provide the learning resources supporting on-campus classes. The Discussion Forums on the Moodle allow group study, discussions and presentations enhancing the full time on-campus experience. References for directed study material and case study material are posted on the Moodle by module tutors and updated as required. Module Co-ordinators, Course Leader, Postgraduate Programme Leader, Academic and support staff are available on campus for support.
	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including:

University Monitoring of Quality and Standards	• <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	This course will comply with university procedures listed above in relation to monitoring of quality and standards. As a standard practice it will include: • internal and external moderation processes for assessments. • interaction with teaching and learning committee for inputs on improving quality of teaching and learning and, design plans to act on feedback from students. • module coordinators will meet teaching staff to discuss syllabi, teaching and assessment, and scope for further improvement in the module. The School of Computing, Engineering and Technology at Robert Gordon University is a learning affiliate member of the Institute of Engineering and Technology (IET).
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience.

	Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	Staff/Student Liaison The School of Computing, Engineering and Technology operates a Staff Student Liaison Committee for its Postgraduate Engineering Programme. The Committee meets at least once per semester. The Postgraduate Programme Management Team receives comments and suggestions from the Student Liaison Committee as a standing agenda item. Students will be encouraged to elect a class representative from their cohort to communicate the views of the cohort to the Staff Student Liaison Committee. Personal Tutors are also available to receive feedback from students.

Awards

Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Postgraduate Certificate - PgCert	Robotics	Aberdeen, United Kingdom	Exit Award		60
Postgraduate Diploma - PgDip	Robotics	Aberdeen, United Kingdom	Exit Award		120
Master of Science - MSc	Robotics	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate the ability to deal with knowledge that covers and integrates the main areas of robotics including their features, boundaries, terminologies and convention considering topics of artificial intelligence, machine learning, advanced control etc.	SC1	SCQF 11
LLO2	Critically analyse understanding of the principal theories, concepts and principles of robotics.	SC1	SCQF 11
LLO3	Critically analyse understanding of the specialised theories of advanced robotics, advanced control, Artificial intelligence, machine learning, computer vision, innovation and sustainability.	SC1	SCQF 11
LLO4	Critically synthesise awareness of current issues in robotics and specialisations of advanced control, artificial intelligence, machine learning and computer vision.	SC1	SCQF 11
LLO5	Demonstrate the ability to design a robotic problem and solution by using a significant range of principal professional skills, techniques, practices and/or materials associated with robotic systems.	SC2	SCQF 11
LLO6	Solve a robotic problem using a range of specialised skills, techniques, practices considering machine learning algorithms informed by forefront developments.	SC2	SCQF 11
LLO7	Produce analysis of problems and solutions in robotic systems using a range of qualitative and quantitative techniques.	SC2	SCQF 11
LLO8	Solve a significant robotic research project, investigation or development through appropriate planning and execution.	SC2	SCQF 11
LLO9	Create original solution to a robotic problem through group work and or individual project work.	SC2	SCQF 11
LLO10	Prepare oneself working on unpredictable technical, non-technical problems while working on interdisciplinary group project and/or individual project work.	SC2	SCQF 11
LLO11	Deal with critical analysis, evaluation and synthesis of forefront issues in the subject area of robotics, advanced control, artificial intelligence and machine learning.	SC3	SCQF 11
LLO12	Construct new and abstract robotic problems by clearly identifying, conceptualizing and defining the issues.	SC3	SCQF 11
LLO13	Generate original and creative responses to robotic problems and issues.	SC3	SCQF 11
LLO14	Deal with critical review, consolidation and extension of knowledge, skills, practices and thinking in subjects of robotics, control, AI and sustainable enterprise development.	SC3	SCQF 11
LLO15	Deal with complex issues and decision making in absence of complete or consistent data through artificial intelligence and machine learning techniques.	SC3	SCQF 11
LLO16	Appraise innovative sustainable robotic and AI/ML solutions to various audiences using presentations and other ICT tools.	SC4	SCQF 11
LLO17	Generate quality reports and articles on proposed sustainable robotics and AI solutions for intellectual communication with peers, senior colleagues and specialist.	SC4	SCQF 11
LLO18	Produce power point slides, recorded videos, demonstrations etc supporting enhancement of proposed robotic, control and AI/ML solutions during the group project stage and the individual project work.	SC4	SCQF 11
LLO19	Critically evaluate numerical and graphical data generated during application of control and AI/ML techniques in coursework and robotics group and individual projects.	SC4	SCQF 11
LLO20	Make informed judgements and initiatives as a member of the robotics group project team or working on the independent project work.	SC5	SCQF 11

LLO21	Take responsibility for decisions or work done in a robotics group project or individual project work.	SC5	SCQF 11
LLO22	Take responsibility for resources and robotics equipment during project work, or data collected from stakeholder groups for sustainable enterprise development and innovation.	SC5	SCQF 11
LLO23	Evaluate the tasks and processes while working in a peer relationship with specialist practitioners during industrial project placements and group projects.	SC5	SCQF 11
LLO24	Take responsibility for tasks demonstrating leadership and/or initiatives making identifiable contributions while working in group or individual coursework and projects.	SC5	SCQF 11
LLO25	Critically evaluate tasks undertaken by self and group members during project work.	SC5	SCQF 11
LLO26	Make informed judgements managing complex ethical and professional issues while working on coursework for innovation and sustainability and sustainable enterprise development with special focus on robotic applications.	SC5	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7

SC5	Autonomy, accountability and working with others	SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
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Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A: Non Placement Route B: Short Placement Route C: Long Placement

Stage 1 / Semester 1**Full Time On Campus**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	ENM280	Advanced Robotics	September Intake	Somasundar Kannan (sk10513)	Yes	1.0	15	Level 11
Full Time On Campus	CM4126	Computer Vision		Eyad Elyan (ee5413)	Yes	1.0	15	Level 10
Full Time On Campus	CEM103	Embedded Systems and Robotic Software		Somasundar Kannan (sk10513)	Yes	2.0	15	Level 11
Full Time On Campus	ENM286	Interdisciplinary Robotics Group Project	January Intake	Somasundar Kannan (sk10513)	Yes	2.0	15	Level 11
Full Time On Campus	CMM541	Machine Learning for Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 11

Stage 1 / Semester 2**Full Time On Campus**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full								

Time On Campus	ENM260	Advanced Control Systems		Somasundar Kannan (sk10513)	Yes	1.0	15	Level 11
Full Time On Campus	ENM280	Advanced Robotics	January Intake	Somasundar Kannan (sk10513)	Yes	1.0	15	Level 11
Full Time On Campus	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
Full Time On Campus	ENM286	Interdisciplinary Robotics Group Project	September Intake	Somasundar Kannan (sk10513)	Yes	2.0	15	Level 11
Full Time On Campus	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11

Stage 1 / Semester 3**Full Time On Campus**

Route A: Non Placement Route B: Short Placement Route C: Long Placement

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Full Time On Campus	ENM280	Advanced Robotics	September Intake	Somasundar Kannan (sk10513)	Yes	1.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	CM4126	Computer Vision		Eyad Elyan (ee5413)	Yes	1.0	15	Level 10
1	Semester 1	Core	Full Time On Campus	CEM103	Embedded Systems and Robotic Software		Somasundar Kannan (sk10513)	Yes	2.0	15	Level 11
1	Semester 1	Core	Full Time On Campus	ENM286	Interdisciplinary Robotics Group Project	January Intake	Somasundar Kannan	Yes	2.0	15	Level 11

			Campus				(sk10513)				
1	Semester 1	Core	Full Time On Campus	CMM541	Machine Learning for Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM260	Advanced Control Systems		Somasundar Kannan (sk10513)	Yes	1.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM280	Advanced Robotics	January Intake	Somasundar Kannan (sk10513)	Yes	1.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CEM101	Innovation and Sustainability for Global Impact		Ruissein Mahon (rm8781)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	ENM286	Interdisciplinary Robotics Group Project	September Intake	Somasundar Kannan (sk10513)	Yes	2.0	15	Level 11
1	Semester 2	Core	Full Time On Campus	CMM544	Natural Language Processing		Laura Tucker	Yes	2.0	15	Level 11
1	Semester 3	Core	Full Time On Campus	CEM100	MSc Project	Routes A, B, C	Jo-Anne Tait (ensjem)	Yes	1.0	60	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM104	Short Postgraduate Placement	Route B	Mark Bartlett (mb9400)	No	1.0	15	Level 11
1	Semester 3	Elective	Full Time On Campus	CEM105	Year-long Postgraduate Placement	Route C	Mark Bartlett (mb9400)	No	1.0	120	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Sources of Further Information

No sources of further information recorded
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